

LORELEI: LoRA Optimization from Reference Examples for Lively, Expressive Illustration

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1 Objective

The goal of this project was to fine-tune Stable Diffusion 1.5 so that the new token <sks> represents the visual style of the Ghibli image collection. As requested, we adapted both the UNet and the CLIP text encoder using LoRA and evaluated the model using the target prompt “a busy market, in <sks> style.”

The main problem was to learn the Ghibli style without losing the base model’s ability to generate a good market. For this reason, we compared training configurations and experimented with using data generated by the untuned SD-1.5 in training to keep the “memory” of what a market is.

2 Image captioning

In order to preserve the way the prompt shapes the images, we thought it would be a good idea to give the training images some captions. But since we’re lazy students, we let a model do this. At first, we tried BLIP for the task but the captions were too short to capture relevant details and oftentimes they were also plain wrong with no apparent visual reason for it. So, we moved on to the next model and settled on Florence-2, using the <CAPTION> task. This provided better results, although it led to some quirks we didn’t want:

- Any images with a top hat were labelled “Professor layton and the Unwound Future”, which is a Nintendo DS puzzle solving game that has a Ghibli-ish art style. To solve this, we automatically changed these captions to “a scene, in style”. See [here](#)
- It often used adjectives like “anime” to describe e.g. the person in the image or often started with “a painting/drawing/cartoon/picture of ...”. Since we wanted it to learn the class token, we just removed these words / starts of sentences with a regex.

The resulting captions are not perfect, but we think they should be good enough or at least better than nothing. Examples of them can be seen in Figure 1.



Fig. 1. Examples of captions generated with Florence-2 for supplied training images.

3 Custom token and LoRA setup

We added the <sks> token to the CLIP tokenizer as a new token with the embedding initialized from the mean embedding of the phrase “Ghibli style”. This increased the tokenizer vocabulary from 49408 to 49409 entries. During training, only the new embedding row was updated.

The U-Net and text encoder were fine-tuned using LoRA with ranks 16 and 4, respectively, targeting their attention projection layers. The final token embedding and both LoRA adapters were stored together in `lora_out/pytorch_lora_weights.safetensors`.

4 Training data

The main dataset contained all 843 supplied 512x512 images. We additionally generated 60 images using the unmodified Stable Diffusion 1.5 base model. This auxiliary set contained 32 broad market scenes, 12 markets with visible medium-sized characters and 16 people or portrait scenes.

In the end, we assigned an effective weight of 86% to supplied style images, 5% to broad markets, 5% to market-character scenes and 4% to people or portraits. Our thinking

was that this way even for longer runs, where we saw that the model “forgot” what markets are, we got better results on the demo prompt.

5 Training experimentation

We trained several candidates and inspected checkpoints at fixed intervals and then selected among the model candidates by visually inspecting the outputs.

5.1 Supplied images only

In the first training run, we tested using only the 843 supplied images. We thought it learned the colours and textures quite well but the generated markets tended to have simpler layouts, more “blob”-like stalls and simpler faces.

5.2 Self-market preservation

We then trained a different model with an additional 60 images generated by the unmodified SD-1.5 model told to generate markets and faces in Ghibli style (turned out in a more broad anime style without the fine-tune but okay-ish enough for us). Through this, we got better stalls and just in generally busier markets as the prompt wants. It provided the best balance between the supplied style and the content requested by the target prompt.

5.3 Balanced people

We also tried to crop faces and persons from the Ghibli-images and re-caption them to then use them as training samples separately to improve faces. In the end, we abandoned this approach because it made faces appear in the background (e.g. in fruit) and didn’t bring the gains in the “real” faces that we hoped for.

6 Final training configuration

The final model used learning rates of $7e-5$ for the UNet LoRA, $2.5e-6$ for the text-encoder LoRA and $1e-5$ for the custom token embedding. LoRA dropout was 0.05. Training used AdamW, fp16, horizontal flips, batch size 1 and four-step gradient accumulation.

We trained the diffusion model using Min-SNR weighting with a gamma value of 5. To prevent the model from changing too much, we added a preservation loss with a weight of 0.65, which encouraged the adapted model to produce similar predictions to the original model for ordinary captions. We also used a token-anchor loss with a weight of 0.05 to keep the custom token close to its initial meaning.

We used a cosine learning-rate schedule with 50 warmup steps and seed 2202. The schedule was defined for 500 steps. The image quality improved noticeably until approximately step 250 and then largely plateaued, so we chose checkpoint 250 for the final adapter.

Other unmentioned hyperparameters can be found in the command-line invocation of the training script shown in our project README.

7 Results

The final fine-tuned model was evaluated using the required target prompt, ‘a busy market, in <sks> style’. Examples of the

generated images can be found in Figure 2. As can be seen, the overall style of the images matches the Ghibli style. In a few places, the generated images look slightly strange, but the overall style is definitely close to that of the provided images. We think that especially the color scheme is more Ghibli-like than the used by the images in the Assignment pdf.



Fig. 2. Results from the selected step-250 model using the required prompt, seeds 84000 to 84003.